Aarav Mittal

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PROFESSIONAL PROFILE

Aspiring Computer Science Engineer at Manipal Institute of Technology, Bangalore, with strong academic performance (CGPA 9.81) and hands-on experience in Full-Stack Development, Artificial Intelligence, and Game Development. Skilled in building autonomous AI agents, deterministic financial software, RAG systems, and interactive 3D/2D games; proven success in hackathons and technical workshops.

EDUCATION

B.Tech Computer Science and Engineering (CGPA: 9.81) July 2025 – July 2029

Manipal Institute of Technology, Bangalore, India - Relevant Coursework: Data Structures & Algorithms, OOP (Java), Data Visualization, Math

Class 12th – CBSE 2025

SNBP International School, Pune, India

Class 10th – CBSE (Score: 97.4%, Top 1% Cohort Rank) 2023

Delhi Public School, Pune, India

WORK EXPERIENCE

Zensar Technologies, Pune June 2026 – July 2026

AI Developer Intern

- **Built end-to-end RAG applications** integrating enterprise knowledge bases with LLMs, slashing technical documentation search and Q&A.
- **Developed custom MCP servers/clients** in Python for standardized, bidirectional tool invocation and dynamic context injection.
- **Designed autonomous AI agents** leveraging MCP for task-driven workflows.
- **Key Project:** RAG-based AI agent enabling PDF upload with contextual Q&A, citations, summaries, study notes, and practice quizzes.

PROJECTS

Parity – AI Money Tracker & Run-Out Forecast [GitHub Link] [Live App]

- Engineered student personal finance web app (Next.js 16, React 19, TypeScript, Supabase RLS) with a 100% deterministic forecasting engine in minor units (paise) to eliminate floating-point bugs (96 unit tests); integrated conversational expense parsing with Groq Cloud and Gemini.

AI Agent: Multi-Feature PDF Document RAG System [GitHub Link] [Live App]

- Architected an autonomous document intelligence assistant featuring PyPDF ingestion, recursive semantic chunking, ChromaDB vector indexing, and FastMCP tool routing for citation-attributed Q&A, executive summaries, notes, and quiz generation.

AI/ML: Real-Time Face Recognition Software (OpenCV, Python) [GitHub Link]

- Developed a real-time computer vision system using OpenCV and Scikit-Learn for ATL AI Workshop to process live camera feeds for multi-face detection, recognition, and automated guest navigation alerts.

God's Hand – 2D Strategy Game (Game Jam, TechSolstice) *5th Place Winner*

- Won 5th Place in 24-hr game jam adhering to "One Screen" theme; engineered game mechanics balancing village terrorization with population equilibrium, custom physics, and NPC state machines.

Dessert Hunt – 3D Unreal Engine Capstone Project [GitHub Link]

- Developed a 3D exploration capstone game in Unreal Engine 5 with responsive character movement, collision dampening, dynamic lighting, and decoupled modular gameplay loops.

SKILLS

- **Programming and Tools:** Python, TypeScript, JavaScript, C++, C, Java, SQL, Git/GitHub, VS Code, MS Office.
- **AI Development:** GenAI, Agentic AI, RAG, MCP, FastMCP, ChromaDB, FAISS, LangChain, LangGraph, ML (Scikit-Learn, OpenCV).
- **Game Development:** Unreal Engine 5, PyGame, Game Design Patterns.

ACHIEVEMENTS

- **5th Place, Game Jam @ TechSolstice** (MIT Bangalore's flagship tech fest).
- **Academic Excellence:** 1st Year GPA of 9.81 / 10.0 and Top 1% of school in CBSE Class 10th (97.4%).